

Experiment 2: Matrix Operations using NumPy

Python Code:

```
import numpy as np
A = np.array([[2, 4],
              [6, 8]])
B = np.array([[1, 3],
              [5, 7]])
print("Matrix A:")
print(A)
print("\nMatrix B:")
print(B)
print("\nAddition:")
print(A + B)
print("\nSubtraction:")
print(A - B)
print("\nMultiplication:")
print(np.dot(A, B))
print("\nTranspose of A:")
print(A.T)
print("\nInverse of A:")
print(np.linalg.inv(A))
```

Output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
= RESTART: C:/Users/A. MOUNESH/AppData/Local/Programs/Python/Python314/Gen AI 1
PY
Matrix A:
[[2 4]
 [6 8]]

Matrix B:
[[1 3]
 [5 7]]

Addition:
[[ 3  7]
 [11 15]]

Subtraction:
[[1 1]
 [1 1]]

Multiplication:
[[22 34]
 [46 74]]

Transpose of A:
[[2 6]
 [4 8]]

Inverse of A:
[[-1.    0.5 ]
 [ 0.75 -0.25]]
```